CI491: Senior Project I

Customer Requirements Document Submission: Vanguard Centralized Segmentation Service

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1. Project Background

The Vanguard Centralized Client Segmentation Service is designed to streamline and standardize how client segments are identified and delivered across Vanguard's digital platforms. Client segmentation enables personalized experiences, helping clients receive appropriate products, services, content, and recommendations based on their profiles.

Today, determining a client's segment often requires querying multiple applications and services during an interaction. This decentralized approach creates complexity, increases development effort, and introduces dependencies on multiple teams when a business area wants to deliver a new personalized experience. Integration challenges and inconsistent segmentation logic can slow innovation across platforms.

The proposed centralized service will combine relevant client information and apply shared, configurable segmentation rules. Applications will be able to retrieve consistent client classifications through a common API, reducing duplicated logic and supporting consistent personalized experiences. In addition to centralized client segmentation, the system will provide intent classification and sentiment analysis to identify client needs and interaction patterns. A demonstration dashboard will allow stakeholders to visualize segmentation results and evaluate these capabilities through simulated client scenarios.

2. Users

User	Role	Needs
Vanguard Client	Uses digital platforms that consume the segmentation service.	Consistent, appropriate personalized experiences based on client information.
Application Developer	Integrates segmentation capabilities into Vanguard digital applications.	A consistent API to retrieve client segments without independently implementing custom segmentation logic.
Business Analyst / Segmentation Owner	Defines segmentation criteria and business requirements.	Clear, maintainable rules that can be reviewed and updated as business needs change.
System Integrator	Maintains connectivity between mock upstream services and the segmentation service.	Well-defined data contracts and predictable handling of missing or invalid source records.

3. Features to Be Implemented

The system will provide centralized, rule-based client segmentation using synthetic data and mock upstream services, along with intent classification, sentiment analysis, and a demonstration dashboard. The following functional and non-functional requirements define the expected system capabilities and qualities.

3.1 Functional Requirements

ID	Feature	Description	Potential Users
FR-01	Centralized Segmentation API	Provide an API that allows applications to retrieve segmentation results for a specified client ID.	Application Developer
FR-02	Client Data Aggregation	Retrieve and combine client information from the profile, account, engagement, and advice mock services.	Application Developer, System Integrator
FR-03	Rule-Based Segmentation	Evaluate client information against configurable segmentation rules stored separately from application logic.	Business Analyst, Developer
FR-04	Service Tier Classification	Determine a client's service tier based on the combined balances of qualifying open accounts.	Application Developer, Business Analyst
FR-05	Life Stage Classification	Classify clients based on their age calculated from their date of birth.	Application Developer, Business Analyst
FR-06	Advice Classification	Identify whether a client is self-directed, digitally advised, or human-advised based on their advice relationship.	Application Developer, Business Analyst
FR-07	Engagement Classification	Identify digitally active and dormant clients based on login recency.	Application Developer, Business Analyst
FR-08	Opportunity Segmentation	Identify rollover candidates, education savers, and advice prospects using information from multiple sources.	Application Developer, Business Analyst

ID	Feature	Description	Potential Users
FR-09	Multiple Segment Assignment	Support mutually exclusive classifications within designated segment families and overlapping assignments across applicable families.	Application Developer, Business Analyst

FR-10	Missing and Conflicting Data Handling	Detect missing records, null fields, and inconsistent data and apply documented handling policies.	System Integrator, QA Tester
FR-11	Configurable Rule Management	Load and evaluate segmentation definitions from configuration data rather than requiring modifications to core evaluation logic.	Developer, Business Analyst
FR-12	Segmentation Result Metadata	Include the evaluation date and rule-set version with segmentation results.	Application Developer, QA Tester
FR-13	Error Handling	Return clearly defined errors when clients cannot be located or requests cannot be processed.	Application Developer, QA Tester
FR-14	Synthetic Data Integration	Support segmentation using supplied fictional datasets served through mock upstream services.	Developer, QA Tester
FR-15	Automated Segmentation Testing	Validate rule behavior against expected results, including boundary values and edge cases.	QA Tester, Developer
FR-16	Intent Classification	Analyze client messages and requests to identify their intended action or service need and return a classification with a confidence score.	Application Developer, System Integrator
FR-17	Sentiment Analysis	Evaluate client interactions to identify sentiment, including frustration or dissatisfaction, to support personalized service experiences and potential human-agent escalation.	Application Developer, Business Analyst
FR-18	Demonstration Dashboard	Provide a visual interface to simulate client scenarios, display segmentation results, and demonstrate intent classification and sentiment analysis using synthetic data.	Application Developer, QA Tester, Project Stakeholder

Feature Clarifications

Service tiers: Assign exactly one qualifying tier when sufficient account information is available. The highest applicable asset threshold determines the tier.

Life stages: The initial catalogue covers clients aged 22 and above. A documented policy is required for clients under 22 and clients with an unknown date of birth.

Advice classifications: Map upstream values such as DIGITAL_ADVISOR, PERSONAL_ADVISOR, and NONE to standardized self-directed, digitally advised, and

human-advised categories.

Engagement categories: Explicitly handle clients whose last login was 31 to 179 days ago, since they do not qualify for either currently defined engagement segment.

Opportunity classifications: Opportunity segments are not mutually exclusive; a client may qualify for multiple opportunities.

Rule configuration: Rules must be defined separately from core evaluation logic. A graphical rule editor is not required.

3.2 Non-Functional Requirements

These requirements define the consistency, maintainability, reliability, and other qualities expected of the service. Response-time targets remain to be determined during evaluation planning.

ID	Requirement	Description	Potential Users
NFR-01	Consistency	Identical client data, rule versions, policies, and evaluation dates must produce identical segmentation results.	Application Developer, Business Analyst
NFR-02	Maintainability	Segmentation rules must be organized and configurable independently of core service logic.	Application Developer, Business Analyst
NFR-03	Modularity	Data aggregation, validation, rule evaluation, and API functionality should remain separate, maintainable capabilities.	Application Developer, System Integrator
NFR-04	Extensibility	The system should support new segmentation rules and families without major changes to existing capabilities.	Application Developer, Business Analyst
NFR-05	Reliability	The service must handle invalid or missing data without unexpected crashes.	Application Developer, System Integrator
NFR-06	Performance	The service should respond efficiently to segmentation requests. Response-time targets will be determined during evaluation planning.	Application Developer, Vanguard Client
NFR-07	Testability	The system must support automated unit, integration, and end-to-end testing.	QA Tester, Application Developer

NFR-08	Reproducibility	All synthetic data evaluations must use the fixed asOfDate of October 1, 2026, unless explicitly configured otherwise.	QA Tester, Application Developer
NFR-09	Observability	The service should provide meaningful logging and error information to support debugging and testing.	Application Developer, QA Tester, System Administrator (future)
NFR-10	Portability	The system should be containerized so it can run consistently across development and demonstration environments.	Application Developer, System Integrator
NFR-11	Data Integrity	Financial calculations must preserve monetary precision and exclude accounts that do not qualify under the defined policy.	Business Analyst, QA Tester
NFR-12	Explainability	Segment assignments should be traceable to the rules and derived data that produced them.	Business Analyst, Application Developer, QA Tester

4. Features Not to Be Implemented

The following features are excluded from the initial project scope.

Feature	**Reason for Exclusion**
Integration with live Vanguard systems	The project uses synthetic datasets and mock services rather than production infrastructure.
Real client authentication	Real authentication and authorization are explicitly out of scope.
Real client financial data	All client records are fictional and used for development and evaluation purposes only.
Investment or financial recommendations	The service classifies clients; it does not generate investment advice or recommend specific financial products.
Transaction processing	The service will not execute trades, modify accounts, or process financial transactions.
Machine learning-based segmentation	The initial segmentation catalogue uses deterministic, rule-based classification rather than predictive models.
Production deployment	Enterprise-scale rollout, live operational support, and deployment into Vanguard's production environment are not part of the initial deliverable.